Laksh. V. Sodhai
D20A
64.

BDA Assignment

1.) consider a webgraph consisting of 4 Pages: A, B, C, D with

- $A \rightarrow B, C$
- $B \rightarrow C, D$
- $C \rightarrow A$
- $D \rightarrow C$

using a dumping factor $d = 0.85$, compute the Pagerank value Initial Pagerank (equal for all 4 Pages) [Given]

$\therefore PR(A) = PR(B) = PR(C) = PR(D) = \frac{1}{4} = 0.25$

- The Pagerank for Page P_i is calculated using the following

$$PR(P_i) = \frac{1-d}{N} + d \sum_{P_i \in min} \frac{PR(P_i)}{L(P_i)}$$

$d = 0.85$, $\frac{1-d}{N} = \frac{1-0.25}{4} = 0.0375.$

$L(P_i)$ is the number of outbound links from Page P_i

- A links to B, C: $L(A) = 2$
- B links to C, D: $L(B) = 2$
- C links to A: $L(C) = 1$
- D links to C: $L(D) = 1$

Iteration 1: using the value from PR_0, calculating PR_1

- $PR_1(A) = 0.0375 + 0.85 \times \frac{PR_0(C)}{1} = 0.0375 + 0.85 \times 0.25$
  $= 0.2500$

- $PR_1(B) = 0.0375 + 0.85 * \frac{PR_0(A)}{2} = 0.0375 + 0.85 \times 0.125$
  $= 0.1438.$

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$$PR_1(C) = 0.0375 + 0.85 \times \left(\frac{PR_0(A)}{2} + \frac{PR_0(B)}{2} + \frac{PR_0(D)}{1} \right)$$

$$= 0.0375 + 0.85 \times (0.175 + 0.175 + 0.25) = 0.4628.$$

$$PR_1(D) = 0.0375 + 0.85 \times \frac{PR_0(B)}{2} = 0.0375 + 0.85 \times 0.175$$
$$= 0.1438.$$

Iteration 2: using values from PR_1, (calculating (PR_2)).

$$PR_2(A) = 0.0375 + 0.85 \times \frac{PR_1(C)}{1} = 0.0375 + 0.85 \times 0.4623$$
$$= 0.4306.$$

$$PR_2(B) = 0.0375 + 0.85 \times \frac{0.75}{2} = 0.1438.$$

$$PR_2(C) = 0.0375 + 0.85 \times \left(\frac{0.21}{2} + \frac{0.1438}{2} + \frac{0.1438}{1} \right)$$
$$= 0.0375 + 0.85 \times (0.125 + 0.0719 + 0.1438)$$
$$= 0.3270.$$

$$PR_2(D) = 0.0375 + 0.85\,(0.1438/2) = 0.0986.$$

After the two iterations the Pagerank values are

Page	Pagerank
A	0.4306
B.	0.1438
C	0.3270
D	0.9986.

2.) Using the same graph, Perform one iteration of HITS algorithm. To compute hub and authority scores.

→ In HITS,

All authority scores = 1

All hub scores = 1

Auth (A) = Auth (B) = Auth (C) = Auth (D) = 1

and Hub (A) = Hub (B) = Hub (C) = Hub (D) = 1

Calculate Authority score

Authority of a Page = Sum of hub scores of Pages linking to Auth (A) = Hub (C) = 1

Auth (B) = Incoming A = Auth (B) = Hub (A) = 1

Auth (C) = Hub (A) + Hub (B) + Hub (D) = 1+1+1 = 3

Auth (D) = Hub (B) = 1

Normalize = $\sqrt{1^2+1^2+3^2+1^2}$.

Page	Normalized Authority
A	0.289
B	0.289
C	0.366
D	0.289

Calculating the scores

Hub of Page = Sum of authority scores of Pages it links to

Hub (A) → links to B, C.

Hub (A) = Auth (B) + Auth (C) = 1+3 = 4 = 1.155.

Hub (B) = Auth (C) + Auth (D) = 0.866 + 0.289 = 1.155.5

Hub (C) = Auth (A) = 0.289.

Hub (D) = Auth (C) = 0.866

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Normalizing $= \sqrt{(1.155)^2 + (1.155)^2 + (0.289)^2 + (0.866)^2}$
$\simeq 1.893$.

Page	Final Hub Score
A	0.610
B	0.610
C	0.453
D	0.457

Hence C is the best authority and A, B are best hubs.

3.) Compare the results obtained from Pagerank and HITS and comment on which Pages are most important for each method.

→ Pagerank perspective

- It identifies Page A and C as most imp. It rewards Pages that are linked to by other high ranking Pages. Since C is heavily linked to and A is linked by C.

HITS Perspective

- Page C is the clear authority because it is the primary destination for most links in the graph.
- Page A and B are the strong hubs because they point to the best authority (Page C).

Conclusion

PageRank provides a single prestige score, effectively highligthning Page A in later iteration as a beneficary of C's popularity. HITS differentiates the roles, showing Page A as content provider.